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Application of Microtunnelling in Oil & Gas Projects for the Installation of Multiple Pipelines

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Abstract

This paper presents the design and execution of a complex river crossing project involving the microtunnelling installation of three hydrocarbon pipelines—16" NB, 14" NB, and 8" NB—beneath the Raidak-II River in the geologically intricate Dooars region of Eastern India.

These hydrocarbon transportation pipelines own and operated by two National Oil companies viz. Oil India Limited and Indian Oil Corporation limited are critical to energy transportation in the region, with two of them operational for over seven decades. The project was initiated by Oil India Limited as part of a phased conversion of overground pipelines—previously suspended from railway bridges—to underground installations for enhanced safety and reliability. The project involved 5 no. of Microtunnelling in total at different river locations.

The Dooars region, located at the foothills of the Eastern Himalayas and characterized by rolling plains, dense forests, and dynamic river systems, posed significant engineering and environmental challenges. While Horizontal Directional Drilling (HDD) and open-cut methods were employed in most locations for pipeline crossing along the route, geotechnical investigations revealed that HDD was unsuitable at certain sites due to subsurface conditions dominated by boulders, cobbles, and pebbles. Consequently, microtunnelling was adopted as an alternative solution.

The project demanded meticulous planning, advanced hydrological and geotechnical assessments, and the use of innovative engineering techniques. Structural modelling using STAAD.Pro and ANSYS played a pivotal role in optimizing the tunnel and support system design, ensuring structural integrity and minimizing risk. A unique aspect of the project was the simultaneous installation of three pipelines within a single RCC casing, offering a cost-effective and space-efficient solution for bunched pipeline crossings beneath major water bodies.

This paper details the design methodology, execution strategy, and site-specific engineering solutions adopted to overcome challenges such as pipeline alignment, shaft sinking in boulder-rich soils, spatial constraints for welding, and environmental disruptions. The successful completion of this microtunnelling operation underscores the viability of bundled pipeline installations in complex terrains and contributes valuable insights for future infrastructure projects in similar geographies.

Introduction

River crossings for hydrocarbon pipelines are critical infrastructure components that demand high reliability and environmental safety. Traditional HDD methods, open cut methods while effective, may not always be feasible due to geological or spatial constraints.

Microtunnelling, also known as pipe jacking, is a non-disruptive method wherein pipes are installed by hydraulic thrust as a Micro Tunnel Boring Machine (MTBM) excavates the soil at the face. The MTBM is launched from a drive shaft and advanced toward a receiving shaft while concrete jacking pipes are simultaneously pushed into place, forming a continuous underground conduit.

Shaft construction was executed using the well sinking method, employing prefabricated segmental steel wedge-shaped cutting shoes. Concrete of M25 grade was poured, and excavation proceeded from within the cutting shoe, with shaft sinking driven by the self-weight of the cast concrete rings.

Thrust wall within the driving shaft/Pit is a Reinforced Cement Concrete block constructed to provide a reaction against the jacking forces by which pipes shall be jacked. All the shafts will have the provision for construction of thrust wall.

Concrete jacking Pipe manufactured in compliance with BS EN 1916 and IS 458 was used. RCC jacking pipe and joint shall be designed to withstand jacking load applied during installation.

The bentonite Mixture acts as lubricant and supports in the pipe jacking Process. Through injection nozzle in concrete pipe such bentonite mixtures is injected into annulus which is created due to the diameter difference between MTBM cutting wheel and concrete jacking pipe. The resulting upward pressure reduces the pipe friction and subsequently the jacking forces. An automatic bentonite lubrication system (ABLS) shall be installed at an interval of approx. 15 metres. Through PLC controller, the bentonite pressure in the annulus space shall be monitored and based on the set pressure, the system is programmed to automatically inject bentonite.

A guidance system comprising of Laser, Target, HWL module and UNS software module was used. The Laser Geo VL-70 is a laser-based pipe alignment and bore path guidance system that was primarily used in microtunnelling. A precise laser beam is projected from the launching shaft through the bore path. A laser target equipped with a digital camera is mounted on the microtunnelling machine (MTBM) or pipe jacking head. As the MTBM advances, the position of the laser dot on the target is continuously monitored by the camera. Based on this real-time feedback, alignment adjustments are made by the operator to ensure that the designed horizontal and vertical trajectory is maintained with high accuracy.

Design considerations

Scour Depth and Cover Requirements

Scour depth refers to the vertical distance by which a riverbed may be eroded due to the movement of water, particularly during high-flow events such as floods. This erosion is caused by the hydraulic forces exerted by flowing water, which can remove sediment and soil from around and beneath structures like bridge piers, abutments, or buried pipelines.

In the context of pipeline crossings beneath riverbeds, scour depth is a critical design parameter. If a pipeline is not buried deep enough, it may become exposed due to scouring, leading to potential mechanical damage, buoyancy issues, or even rupture—posing serious environmental and safety risks.

To ensure long-term stability and compliance with Indian standards such as PNGRB T4S and OISD-141, the tunnel for the hydrocarbon pipelines under the Raidak-II River was designed with a minimum cover of 2.5 meters below the predicted maximum scour depth. This conservative approach was based on the analysis of historical hydrological data, which helped estimate the maximum depth to which the riverbed could potentially erode over time. By maintaining this cover, the design ensures that the pipelines remain securely buried and protected even under extreme hydraulic conditions.

Geotechnical and Topographical Surveys

A comprehensive geotechnical survey was conducted to support the design and execution of the pipeline crossing beneath the Raidak-II River. This survey included soil and rock sampling, Standard Penetration Tests (SPT), Cone Penetration Tests (CPT), groundwater table measurements, and geophysical investigations such as seismic refraction and electrical resistivity profiling. Laboratory testing of collected samples provided critical data on soil strength, permeability, and consolidation characteristics. These findings guided the selection of appropriate microtunnelling tools and informed the design of the tunnel alignment and shaft depths. In conjunction with topographical and hydrographic surveys, the geotechnical data ensured that the pipeline maintained a consistent cover below the predicted scour depth across the entire 368-meter crossing. This approach helped mitigate risks associated with soil instability, groundwater intrusion, and potential exposure due to riverbed erosion, thereby ensuring structural integrity and compliance with Indian standards.

Tunnel and Shaft Design

Tunnel Diameter. The selection of the microtunnel diameter is a critical design consideration, particularly when accommodating multiple pipelines within a single casing. In this project, a 1600 mm internal diameter (ID) reinforced cement concrete (RCC) casing pipe was chosen to house three hydrocarbon pipelines of 16", 14", and 8" nominal bore (NB). The determination of this diameter was based on several interrelated factors.

Foremost among these is the cumulative outer diameter of the carrier pipelines, including allowances for external coatings, insulation, and spacers. Adequate clearance between the pipelines and the casing wall is essential to facilitate safe installation, alignment adjustments, and the placement of grout or other filler materials in the annular space. The design must also consider the minimum bending radius of the pipelines to prevent mechanical stress during insertion.

Furthermore, the casing must be structurally capable of withstanding external loads, including overburden pressure, hydrostatic forces, and potential surface loads from vehicular or construction activity. The selected diameter must also be compatible with the capabilities of the microtunnelling equipment, particularly in terms of jacking force and guidance control. These considerations collectively inform the tunnel profile and ensure the safe, efficient, and regulation-compliant installation of multiple pipelines within a single microtunnel.

Shaft Configuration. Circular shafts were selected for both launching and receiving operations due to their inherent structural efficiency and ease of sinking in varied ground conditions. The launching shaft was designed with an internal diameter of 8 meters, while the receiving shaft measured 7 meters. Circular geometry offers uniform distribution of earth and hydrostatic pressures, reducing the risk of localized stress concentrations and simplifying construction logistics.

To ensure structural integrity under anticipated loads—including soil pressure, groundwater forces, and construction equipment—STAAD modelling was employed for detailed structural analysis and design. The modelling facilitated accurate simulation of load conditions, material behaviour, and reinforcement requirements, enabling the design of safe and durable shaft structures compliant with relevant standards. This approach ensured that both shafts could withstand operational stresses during microtunnelling and long-term environmental exposure.

Pipeline Support and Spacing

Use of Metallic Roller Spacers for Smooth Installation. Carrier pipelines were supported using metallic roller spacers strategically placed along the alignment. These rollers facilitated low-friction movement of the pipe during insertion, minimizing mechanical stress and ensuring a smooth and controlled installation process, particularly through microtunnelling or HDD shafts.

Load Isolation Between Adjacent Pipelines. The support system was specifically designed to prevent load transfer between parallel or bundled pipelines. By isolating individual carrier pipes mechanically, the system reduced the risk of differential settlement, stress concentration, and possible misalignment, thereby preserving the integrity of each pipeline during both installation and operation.

Provision of Adequate Working Clearance. Sufficient spacing between the pipes and between the pipes and shaft walls was maintained to facilitate safe and effective execution of field welding and Non-Destructive Testing (NDT) activities. This clearance allowed technicians to operate welding torches, scanners, and ultrasonic devices without obstruction, ensuring high- quality joint integrity and compliance with inspection standards.

Buoyant Force Resistance in Case of Water Ingress. The roller spacer assemblies were structurally designed to withstand potential buoyant forces arising from accidental or unavoidable groundwater ingress into the microtunnelling or casing annulus. These rollers-maintained pipeline alignment and stability under submerged or partially submerged conditions, preventing pipe uplift or floating which could lead to misalignment or damage.

Temporary Cathodic Protection

The design of the Temporary Cathodic Protection (TCP) system for the microtunnelling (MT) section is based on site-specific conditions. Key parameters influencing the selection of the cathodic protection system include the pipeline geometry, adjacent structures, soil resistivity, spatial constraints, construction methodology, and the presence of stray currents from nearby installations.

Additional considerations such as accessibility of the pipeline route, optimal positioning of cathodic protection components (e.g., anodes, test stations), and overall economic feasibility further define the limitations and scope of the TCP design.

For the MT section at Raidak-II, a sacrificial anode-based TCP system has been adopted using high-potential, pre-packed magnesium anodes (10 kg each). As a minimum requirement, four such anodes are proposed to be installed—two at each end (Shaft 1 and Shaft 2) of the microtunnelling segment. This ensures full and adequate protection, irrespective of design calculations that may suggest a lower number of anodes.

Test stations are planned at both ends of the MT section to facilitate pipe-to-soil potential measurements. Additionally, a test station was installed near the insulating joint (IJ), located underground in a pit at a nearby sectionalizing valve (SV) station. To safeguard the insulating joint against high-voltage surges, a 100 kA-rated surge diverter was installed across the IJ. This provision ensures the system's resilience against transient overvoltage events.

This TCP design approach ensured effective and reliable corrosion protection for the critical microtunnelling segments during the construction and commissioning phases.

Table 1—Data table for RCC construction

Sl. No.	Description	Specification
1.	Size of the Jacking Pipe	1600 mm ID, 1930 mm OD in RCC jacking Pipes
2.	Design Drive length	380 meters
3.	Shaft Sizes	Launching Shaft S1 (8-meter ID / 9.6-meter OD).Receiving Shaft S2 (7-meter ID / 9.6-meter OD).
4.	Shaft Depth	16 meters from Ground level.
5.	Inclination angle	Nil
6.	Jacking pipe length	2.5 meter
7.	Jacking Pipe weight	5.5-6.0 Ton

Table 2—MTBM Machine and Controller Details

Sl. No.	Description	Specification
1.	Cutting Head type	Rock
2.	Outside Dia.	1970 mm
3.	Maximum Torque	360 KNM
4.	Maximum Speed	7.7 RPM
5.	No. of Steering Cylinders:	4 Nos
6.	Slurry Circuit	DN 150mm
7.	Hydraulic Pump	110 KW
8.	Maximum jacking force (F):	8548 N
9.	Maximum pressure:	500 bars
10.	First stroke:	1690 mm
11.	Second stroke:	1830 mm
12.	No. of cylinder:	2 Nos

Structural and FEA Analysis

- RCC jacking pipes were checked for jacking and superimposed loads.
- A thrust wall was constructed to resist jacking forces.
- ANSYS-based FEA was used to validate support design, bearing loads, and buoyancy resistance.

Execution challenges and solutions

During the execution of the microtunnelling project involving three parallel hydrocarbon pipelines, several technical and environmental challenges were encountered. Each challenge required site-specific engineering solutions to ensure the successful completion of the pipeline installation. The major issues faced, and their corresponding mitigation strategies are elaborated below:

Balancing and Alignment of Multiple Pipelines

One of the foremost challenges was maintaining the structural balance and alignment of the three carrier pipelines during installation. The unequal outer diameters and wall thicknesses of the pipelines led to variations in individual pipe weights, which in turn caused uneven load distribution within the tunnel. This imbalance posed risks of differential settlement, rotational shift, and misalignment during pipe pushing operations.

To overcome this, custom-designed roller support assemblies were fabricated to accommodate the specific weight profile of each pipe. These supports ensured uniform distribution of vertical and lateral loads and helped maintain precise alignment of all three pipelines throughout the tunnel length. The support system was also designed to isolate the load transfer between the pipelines, thereby reducing structural interdependence.

Shaft Sinking in Boulder-Rich Soil Conditions

The project site was characterized by geotechnical complexities, particularly during shaft construction, where the presence of cobbles and large boulders significantly hindered excavation activities. Standard excavation methods proved ineffective, leading to delays and increased mechanical wear.

To facilitate shaft sinking under these adverse conditions, additional dead weights in the form of pre-cast concrete blocks were deployed to increase the downward force on the shaft cutting edge. In addition, Reinforced Cement Concrete (RCC) lift segments were introduced at regular intervals to provide structural support to the shaft wall, maintain verticality, and aid in the progressive advancement of the sinking operation. These measures collectively helped in overcoming soil resistance and achieving the desired shaft depth within tolerances.

Space Constraints for Welding Operations

Given the confined geometry of the microtunnelling shafts and limited access inside the casing pipe, space availability for simultaneous welding of all three pipelines was severely restricted. Direct alignment of girth weld joints across all three pipelines would have led to obstruction and inefficiencies in executing welding and inspection tasks.

This constraint was mitigated by staggering the joint locations of the three pipelines longitudinally. The staggered arrangement ensured that welding crews could operate on each pipe without interference, while also providing adequate space for NDT activities such as ultrasonic testing and radiography. This approach maintained productivity and ensured adherence to safety and quality protocols.

Environmental Disruptions and Accessibility

The project timeline was adversely affected by seasonal monsoon flooding and elevated groundwater levels, which led to temporary suspension of activities at multiple sites. In some locations, approach roads to the shaft entry points were washed away, thereby restricting the movement of personnel, machinery, and materials.

To mitigate these disruptions, temporary access routes were reconstructed using compacted earth and granular sub-base materials, often supplemented with geotextile reinforcement for added stability. Dewatering systems and embankment reinforcements were also implemented to manage water ingress and protect working areas. These remedial actions enabled safe resumption of operations and limited the overall delay in project progress.

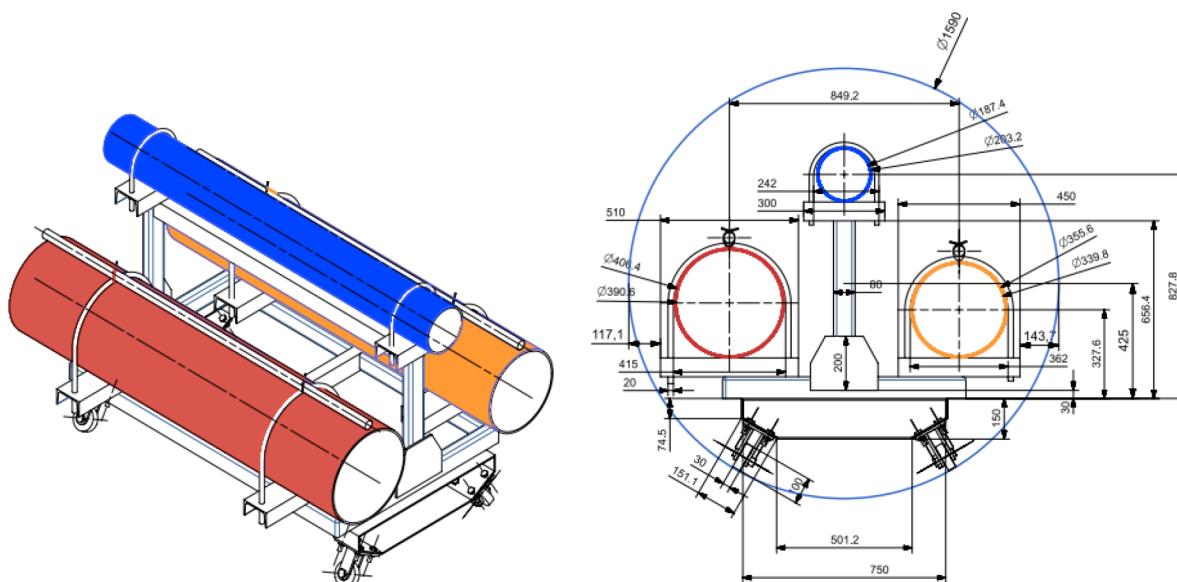


Figure 1—Arrangement of three pipelines inside the Microtunnel

Discussion & Results

The microtunnelling operation was successfully completed with the installation of all three hydrocarbon pipelines within the Reinforced Cement Concrete (RCC) casing. The execution adhered strictly to the design specifications, ensuring structural integrity, environmental compliance, and operational safety across all phases of the project.

The final pipeline configuration demonstrated excellent alignment and stability, with no observable signs of settlement or deformation post-installation. The RCC casing provided robust protection against external loads and environmental stressors, thereby enhancing the long-term durability of the pipeline system.

A key factor contributing to the success of the project was the integration of advanced structural and geotechnical modelling tools during the design phase. Software platforms such as STAAD.Pro and ANSYS were employed to simulate load conditions, assess stress distribution, and optimize the support systems within the tunnel. These tools enabled the engineering team to:

- a. Evaluate the interaction between the pipelines and the surrounding tunnel structure.
- b. Predict and mitigate risks related to differential loading and soil-structure interaction.
- c. Design custom support assemblies and shaft components with high precision.
- d. Validate the structural performance of the RCC casing under dynamic and static loads.

The use of finite element analysis (FEA) and structural modelling not only improved design accuracy but also facilitated proactive decision-making during construction. This approach minimized rework, reduced material wastage, and ensured compliance with safety and quality standards.

Overall, the project stands as a testament to the effectiveness of integrated engineering design and execution strategies in overcoming complex subsurface construction challenges.

Conclusion

The successful execution of this project underscores the technical viability and reliability of microtunnelling as a superior alternative to Horizontal Directional Drilling (HDD) for complex, multi-pipeline river crossings, particularly in geologically and environmentally challenging regions. By addressing limitations often encountered in HDD—such as subsoil instability, limited steering control, and higher risk of frac-outs—microtunnelling proved to be a controlled, trenchless solution capable of delivering precise alignments with minimal ecological disturbance.

A critical factor in the project's success was the integrated engineering approach, which combined comprehensive geotechnical investigations, advanced structural modelling, and the deployment of custom-designed support systems. This integration enabled engineers to proactively manage risks associated with uneven load distribution, buoyant forces, space constraints, and subsurface variability. The use of metallic roller spacers, balanced support configurations, and shaft sinking aids such as RCC lifts and additional weights enhanced both the structural integrity and the installation precision of the carrier pipelines.

The project also demonstrated a high degree of adaptability in overcoming execution challenges, including boulder-laden strata, restricted welding space, and adverse environmental conditions like monsoonal flooding and access disruption. These challenges were addressed through site-responsive engineering solutions, including staggered welding joints, modular shaft design, and temporary road restoration, enabling safe and uninterrupted progress under difficult field conditions.

Importantly, the project also validated the cost-effectiveness of microtunnelling for multi-pipeline installations in critical crossings. Although initial capital expenditure may appear higher than conventional open-cut or HDD methods, the long-term benefits in terms of reduced rework, minimal environmental restoration costs, lower risk of failure, and extended service life of the pipeline system contribute to significant overall cost savings. The minimized downtime, reduced land acquisition, and lower socio-

environmental impact further enhance its economic attractiveness for pipeline operators and project developers.

In conclusion, this case study not only demonstrates the feasibility and cost-effectiveness of installing multiple hydrocarbon pipelines via microtunnelling but also provides a replicable model for future projects requiring high precision, resilience, and sustainability under complex ground and environmental conditions. It reinforces the importance of early-stage multidisciplinary planning, innovative construction practices, and holistic risk and cost management to ensure technically sound, economically viable, and environmentally responsible pipeline infrastructure development.

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